# Bacterial cyclooxygenase homologs might be associated with multicellularity

Poster talk video: youtu.be/iK9nVoEUfkE

Poster talk video: youtu.be/iK9nVoEUfkE

**Georgy Kurakin[1] and Nadezhda Potapova[2]**

[1]Pirogov Russian National Research Medical University, *Moscow, Russia*, ***email:*** *georgykurakin@gmail.com*

[2]Kharkevich Institute of Information Transmission Problems RAS, *Moscow, Russia*, ***email:*** *nadezhdalpotapova@gmail.com*

## What did we know?

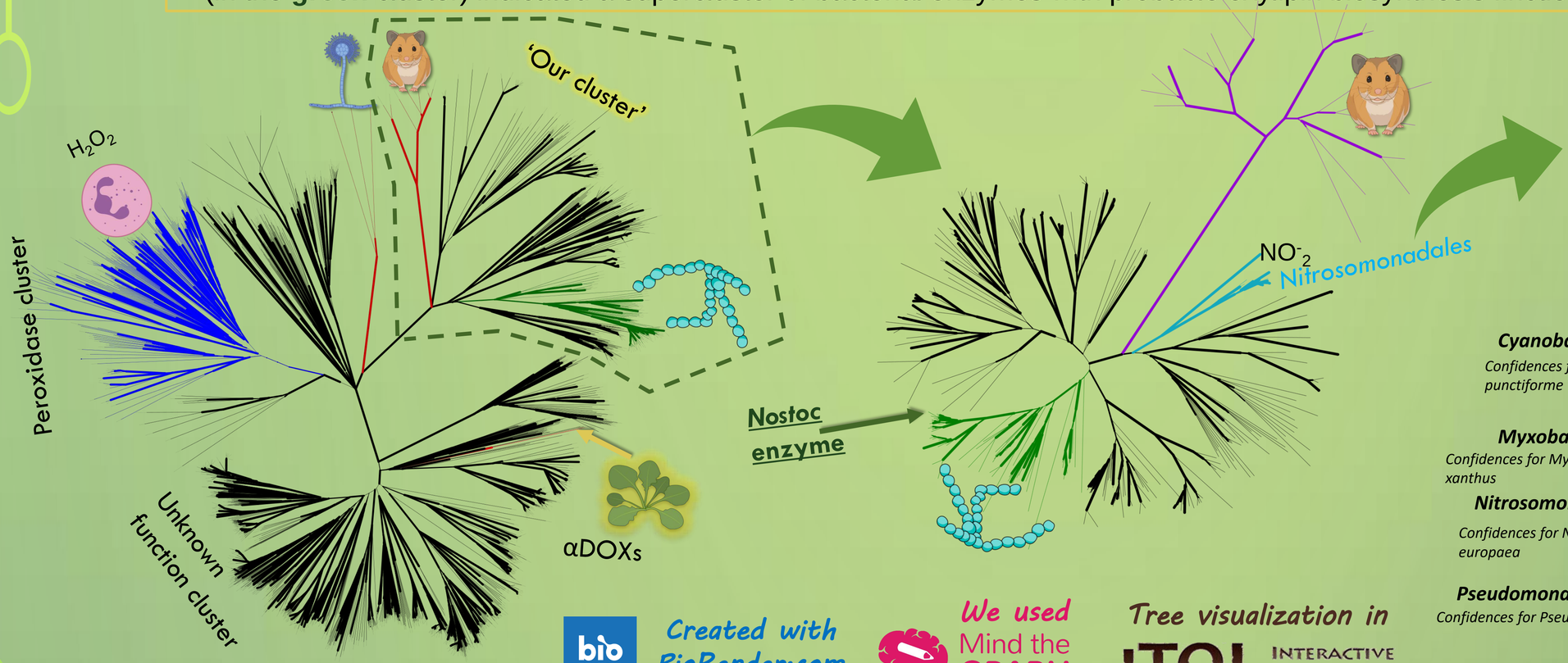
- Oxylipins are **almost universal** signalling compounds across all multicellular eukaryotes.
- Oxylipin biosynthesis is provided by homologous **lipoxygenase**-like and **cyclooxygenase**-like enzymes in different kingdoms.

## What was recently found?

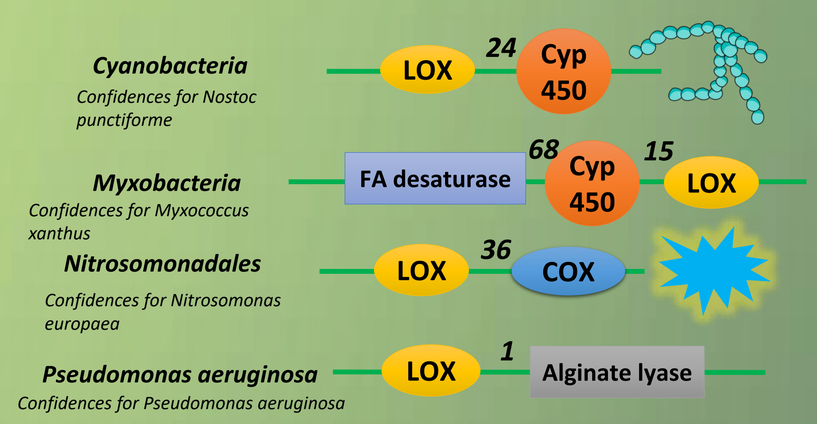
- Lipoxygenases (LOXs) could be evolutionarily associated with multicellularity (**Paper 1**, **Paper 2**).
- Some bacteria might use LOXs for **cross-kingdom pathogenicity** and **immune response evasion**.

## *What about cyclooxygenase-like (COX-like) enzymes? Are they associated with multicellularity?*

- We constructed a series of phylogenetic trees of bacterial cyclooxygenase homologs, animal cyclooxygenases (COXs), plant α-dioxygenases (αDOXs), and fungal COX homologs.
- Some of these enzymes perform **different function**, but a **biochemically characterized** *Nostoc* COX-like enzyme with a lipoxygenase activity (in the **green** cluster) indicated a supercluster of bacterial enzymes with probable oxylipin biosynthesis finction. We called it **"Our cluster".**




Indirect **extra-phylogenetic confirmation** of the close proximity of animal COX and nitrosomonadal homologs: LOX and COX-like protein **in one operon in *Nitrosomonadales*** and their functional synergy in animals. (**OperonDB** data).




Created with BioRender.com

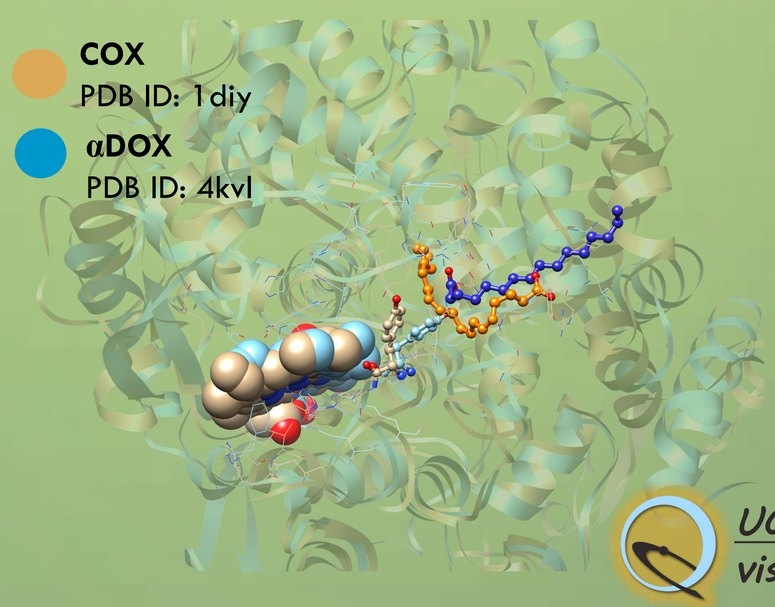
We used Mind the GRAPH

Tree visualization in iTOL Interactive Tree Of Life

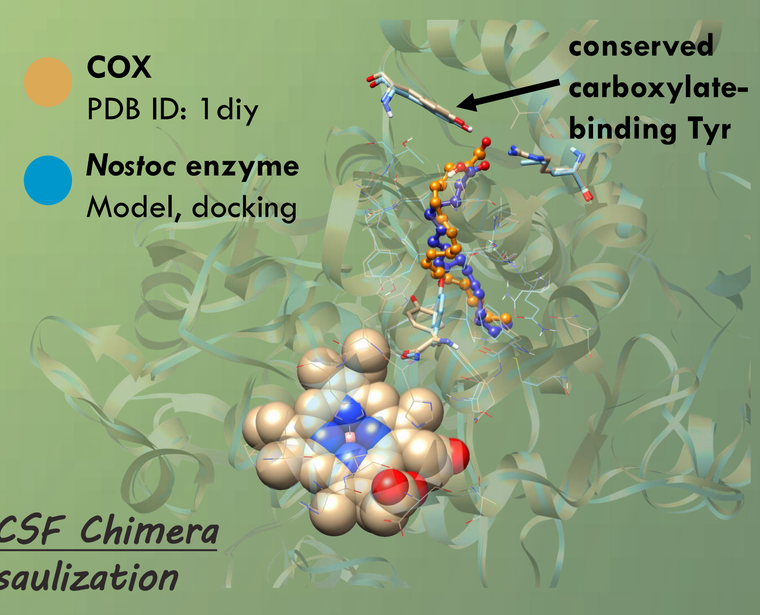
**Substrate-binding residues** are **conserved** between COXs ans *Nostoc* COX-like protein, but **evolved independently** in αDOXs and fungal COX-like enzymes.

| COX1 ovine | ppoC Asp. Fum. | αDOX | COX-like *Nostoc* | Myeloperoxidase |
|---|---|---|---|---|
| V116 | R | V | L | |
| R120 ← | L | L | L | |
| F205 | I | M | F | |
| F209 | I | W | F | |
| Y348 | Y | W | Y | |
| V349 | L | T | I | |
| L352 | I | L | I | |
| Y355 ← | L | T | Y | |
| N375 | N | Y | N | |
| I377 | C | L | M | |
| **F381** | **F** | **F** | **F** | F |
| **Y385** | **Y** | **Y** | **Y** | F |
| W387 | W | M | W | Y |
| M522 | I | A | V | |
| I523 | A | F | A | |
| G526 | Y | F | I | |
| A527 | T | L | G | |
| S530 | A | A | A | |
| G533 | L | R | Q | |
| L534 | V | L | A | |
| *E-val* | *~$10^{-4}$* | *N.S.S.* | *~$10^{-11}$* | |

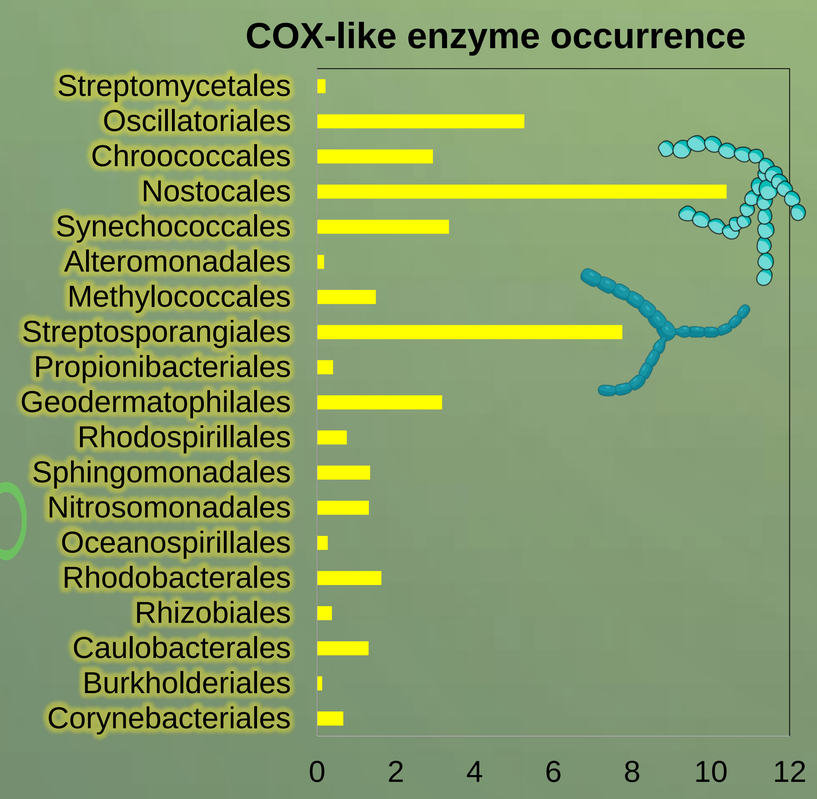
This is confirmed by the **substrate placement** in COX and αDOX — it is **almost opposite** (PDB IDs: **1diy**, **4kvl**, **UCSF Chimera** model).




In contrast, the docking results suggest that the **bioclemically characterized** ***Nostoc*** **enzyme** might bind the substrate **like COX** (**AutoDock Vina** docking, **UCSF Chimera** model).




UCSF Chimera visaulization




- We have obtained the indirect evidence **that "our cluster" proteins are oxylipin-producing enzymes** of bacteria closely related to our **COXs.**
- The occurrence of "our cluster" enzymes (***at the left***) is at the highest in **filamentous cyanobacteria** and ***Streptosporangiales*** **actinobacteria**. They are **multicellular**.
- This leads to a **conclusion** that COX-like enzymes in bacteria are **statistically associated with multicellularity** (like LOXs).
- They might be involved in **oxylipin cell-to-cell signalling** for maintaining **complex structure**, like human COXs and plant αDOXs.

Find out more in our **slide deck** on **ResearchGate**:
*Kurakin, G.F., Potapova, N.A. (2023)* ***Insights into the cyclooxygenase phylogeny****. Unpublished.*
*https://doi.org/10.13140/RG.2.2.35390.46402*
**Methods** and **full references** are included.